

Self-Propelled Arm System



Project presentation

Filip Chekalkin, Huaijing Hou, Spaska Janeva, Kamen Kanev,
Giorgi Khundadze, Xiaosheng Tan, Stoyan Vlaev, Weiye Yuan

Embedded Systems, Cyber-Physical Systems and Robotics
Technical University of Munich, Campus Heilbronn

Introduction and Outline

Overview

System summary

■ Autonomous can collection robot

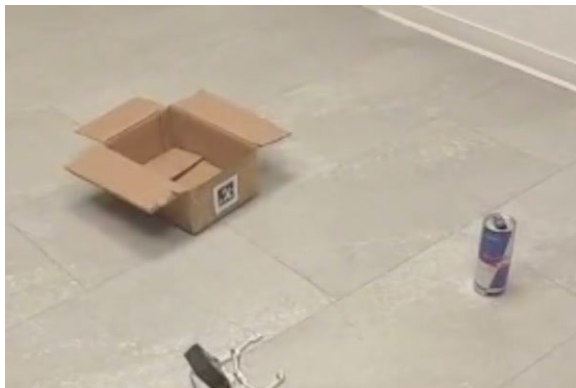
Tracked indoor platform with a four link arm and a parallel gripper

■ A single RGB camera as the only sensor

No depth sensor, no range finder, no wheel encoders, no force sensing

■ Vision in the loop at every stage

Search, approach, grasp and delivery are decided from the image



Outline



Problem and platform

Task definition and hardware constraints



Architecture and control

System components, state machine, grasping



Perception and demonstration

Detection results and a complete run on hardware



Experimental results

Detector accuracy and mission timing



Limitations and conclusion

Open issues, summary and future work

Problem Statement

Objective and constraints

- Drink cans accumulate in indoor public spaces
- Collection is repetitive, low risk manual work
- The task combines perception, navigation, decision making and manipulation



Detection

Rotate in place until the detector accepts a target



Approach

Steer on the image error, stop on apparent size



Grasp

Lower the arm, push, close, lift



Delivery

Dock on the AprilTag and release above the bin

Sensor configuration

■ Single RGB camera

■ No depth sensing

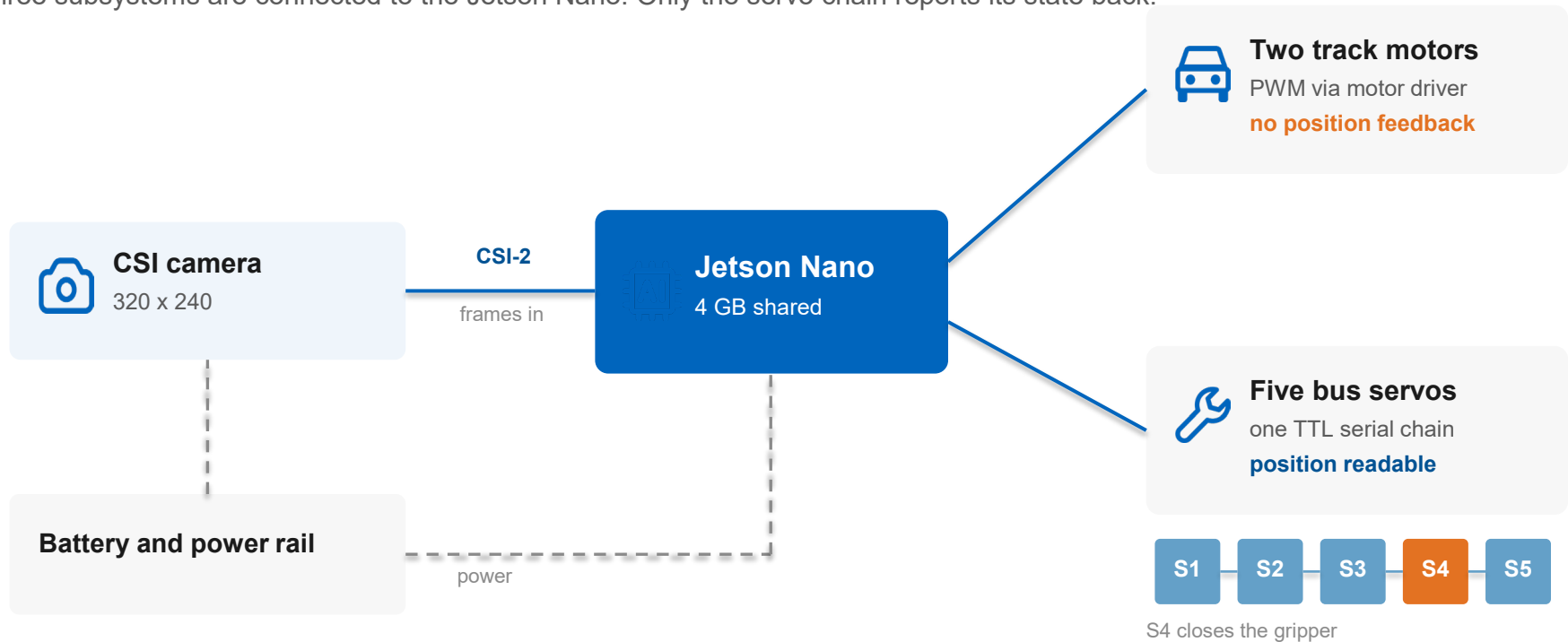
■ No wheel encoders

■ No force sensing

System Architecture

Components and software framework

Three subsystems are connected to the Jetson Nano. Only the servo chain reports its state back.

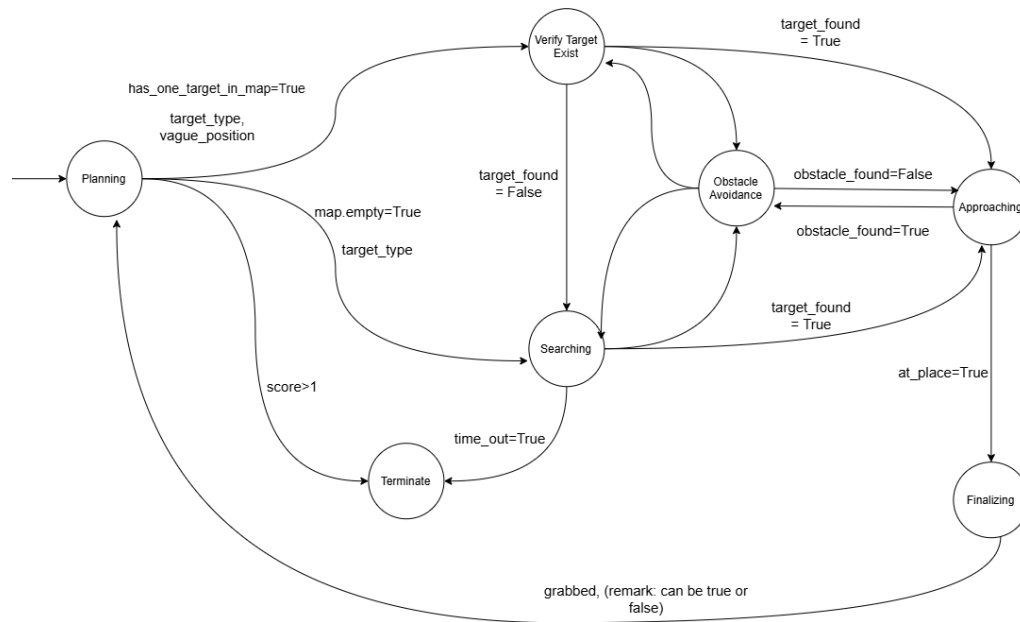


Software stack



State Machine and Control Logic

Mission control



Design level state graph. Transition guards are fields of a single shared context object.

Vision driven states

■ Searching

Rotate until a target is accepted or the timeout expires

■ Aligning

Turn to tolerance, abort if the target is lost

■ Approaching

Steer or drive forward, stop on apparent size

■ Final verification

Tighter tolerance over consecutive frames

$$e_x = (x_c - W/2) / W$$

- Range -0.5 to 0.5, negative to the left of centre
- Stop thresholds: 0.36 box height, 0.15 tag height

- Transitions are held in an explicit table, so an undeclared state and event pair raises an error
- Finalising requires a stable target, a condition map based navigation never reaches, so dead reckoning triggers no action

Grasping Strategy

Manipulation sequence

Fixed poses — no inverse kinematics

1

Wait

2.0 s settle

2

Lower

arm_down pose

3

Push

Slow forward drive

4

Hold

5.0 s lock

5

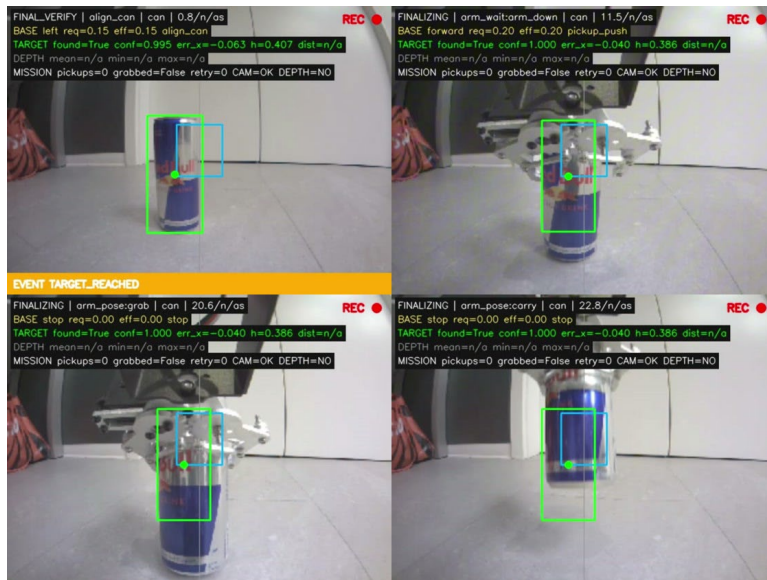
Close

Grab pose

6

Lift

Carry pose



Why push, not reach

- Short arm, imprecise gripper
- Base drives reliably
- Can driven into open gripper
- Coarse motion beats fine manipulation

Servo feedback

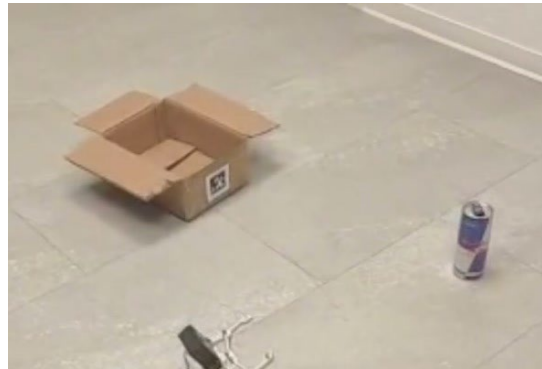
- Polled until readings stabilise
- Push blocked until settled
- 4.94 s settle · 7.0 s push · 5.0 s hold

Verify · push · close · carry

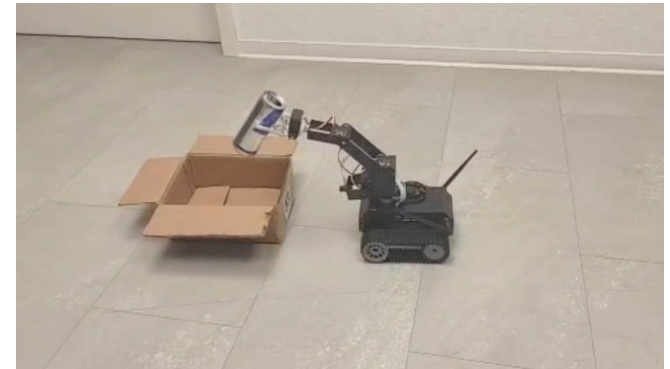
Experimental Setup



Platform with grasped can



Arena: can, AprilTag bin



Release above bin

73.8 s

Full run

1

Camera only

98

Can detections

41 / 59

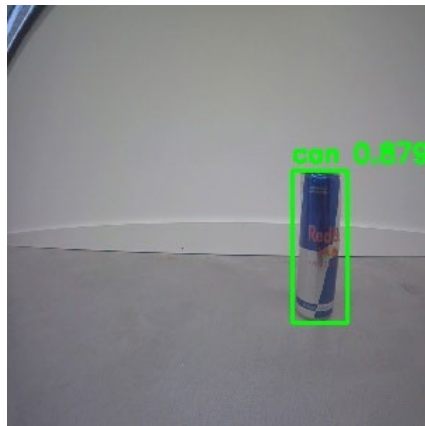
Tag reads

Video



Object and Marker Detection

Two detectors, every controller input



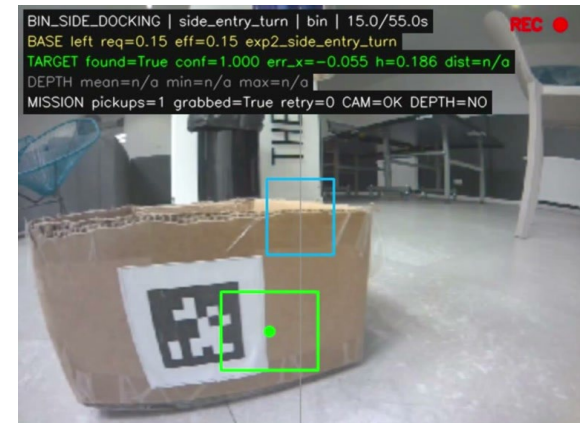
Detector output

SSD-MobileNet-V1 · 300 × 300



Can detection

error -0.119 · height 0.255



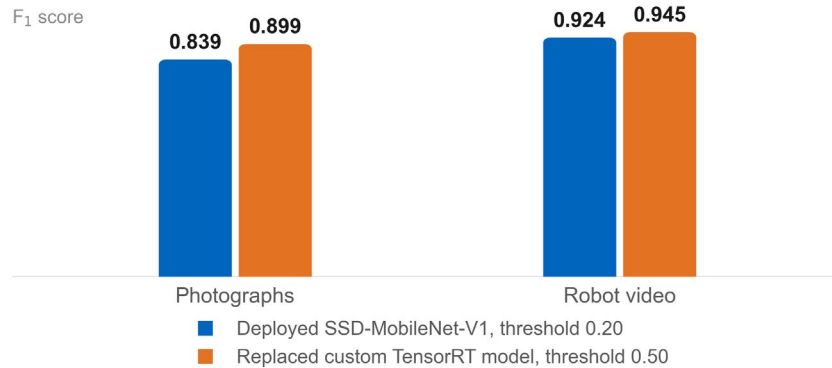
Tag detection

tag36h11 id 0 · height 0.186

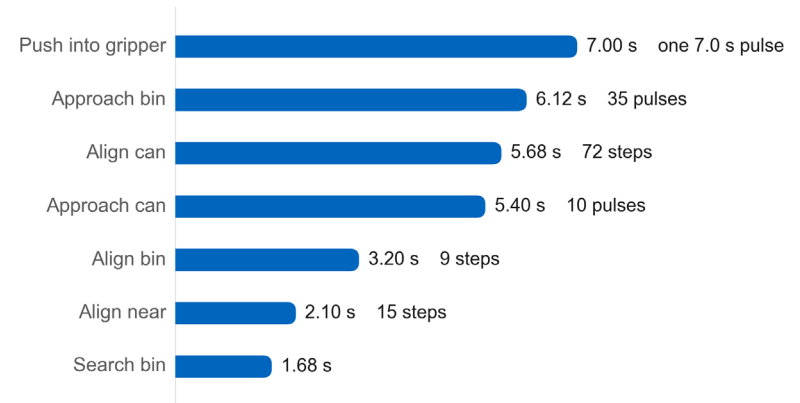


Experimental Results

Detector accuracy, F1



Base motion per phase



73.8 s

Full mission

31.2 s

Base motion, 42 %

11.9 s

Deliberate holds

4.4×10^{-7}

PyTorch vs. ONNX

Accuracy vs. deployability

- -6 F1 photos, -2 video
- No custom post-processing
- Native NVIDIA path

Caveat: archived run

- Earlier prototype, TensorRT detector
- Legacy thresholds 0.400 / 0.500
- Counts and timings exact

Limitations and Challenges

Evaluation



Depth excluded

- 0.3 m general scenes
- 1 m white surfaces
- Detector reaches 3 m

Removed from control path



Pose drifts

- Left / right: 1–3 rad/s
- Modelled 0.471 / 0.942
- Error exceeds model

AprilTag: only anchor



Silent failure

- Verification disabled
- Pickup always recorded
- Empty gripper reports success

First fix: verification flag

Missing capability: relocalisation

Principal challenges

■ Open-loop turning unreliable

Pulses + visual correction

■ No gripper feedback

Full and empty identical

■ 10–15 fps

Pulsed motion, low speeds

■ Model deployment

Accuracy traded for reliability

■ Depth underperformed

Unreliable at useful ranges

■ Grasp timing

Polling beat fixed delay

Conclusion and Future Work

Summary

Summary

- Mobile manipulator, single camera
- Closed on appearance, not geometry
- Errors degrade, never break
- Drifting map lengthens search
- Asymmetric turning needs corrections

Next steps

- **Enable grasp verification**
- **Calibrate turn constants**

Sensors worth adding



Forward distance sensor

First metric measurement



Gripper force sensing

Fixes empty gripper



Wheel odometry

Map becomes measurement



More room tags

Cheap relocalisation anchors